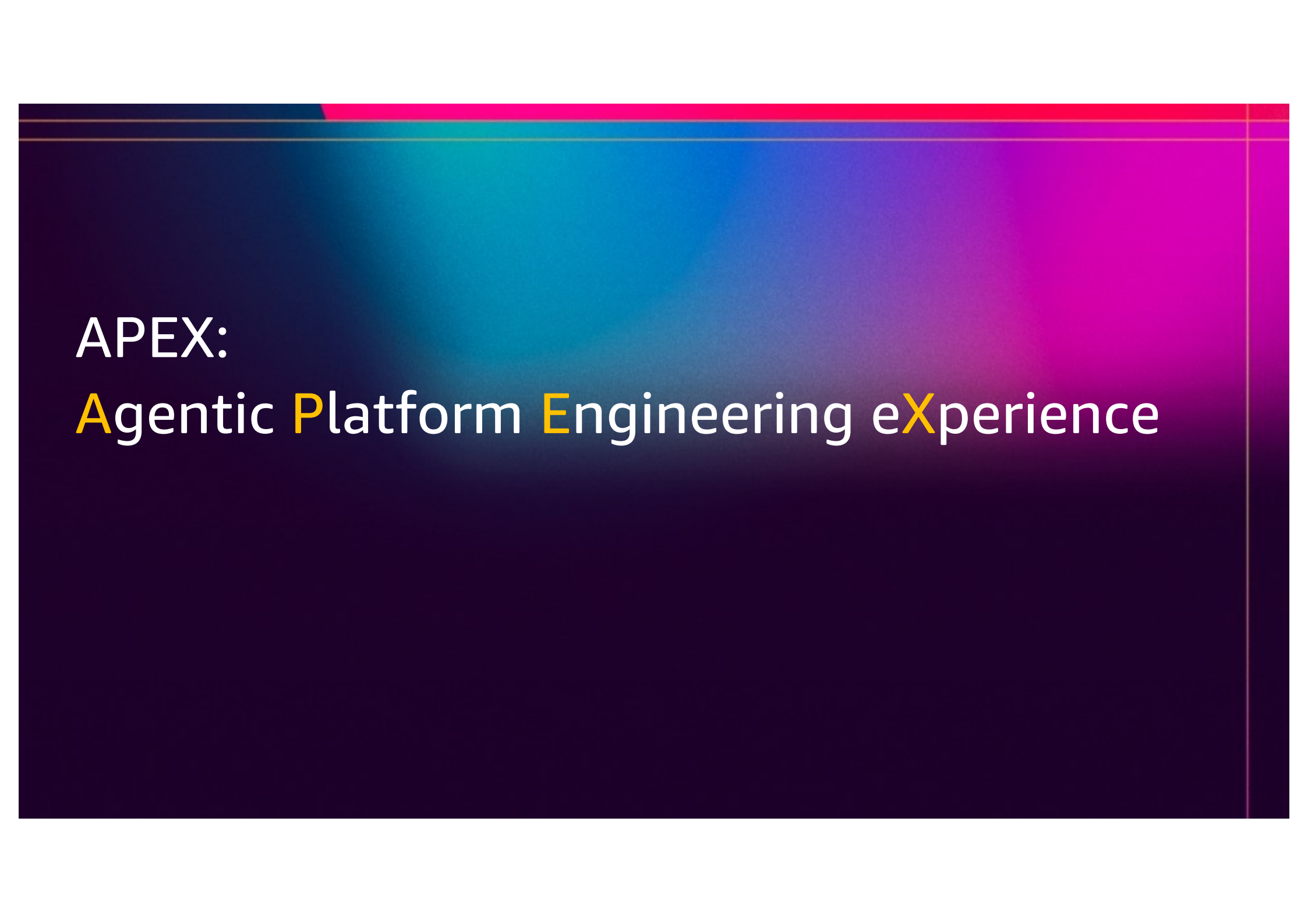




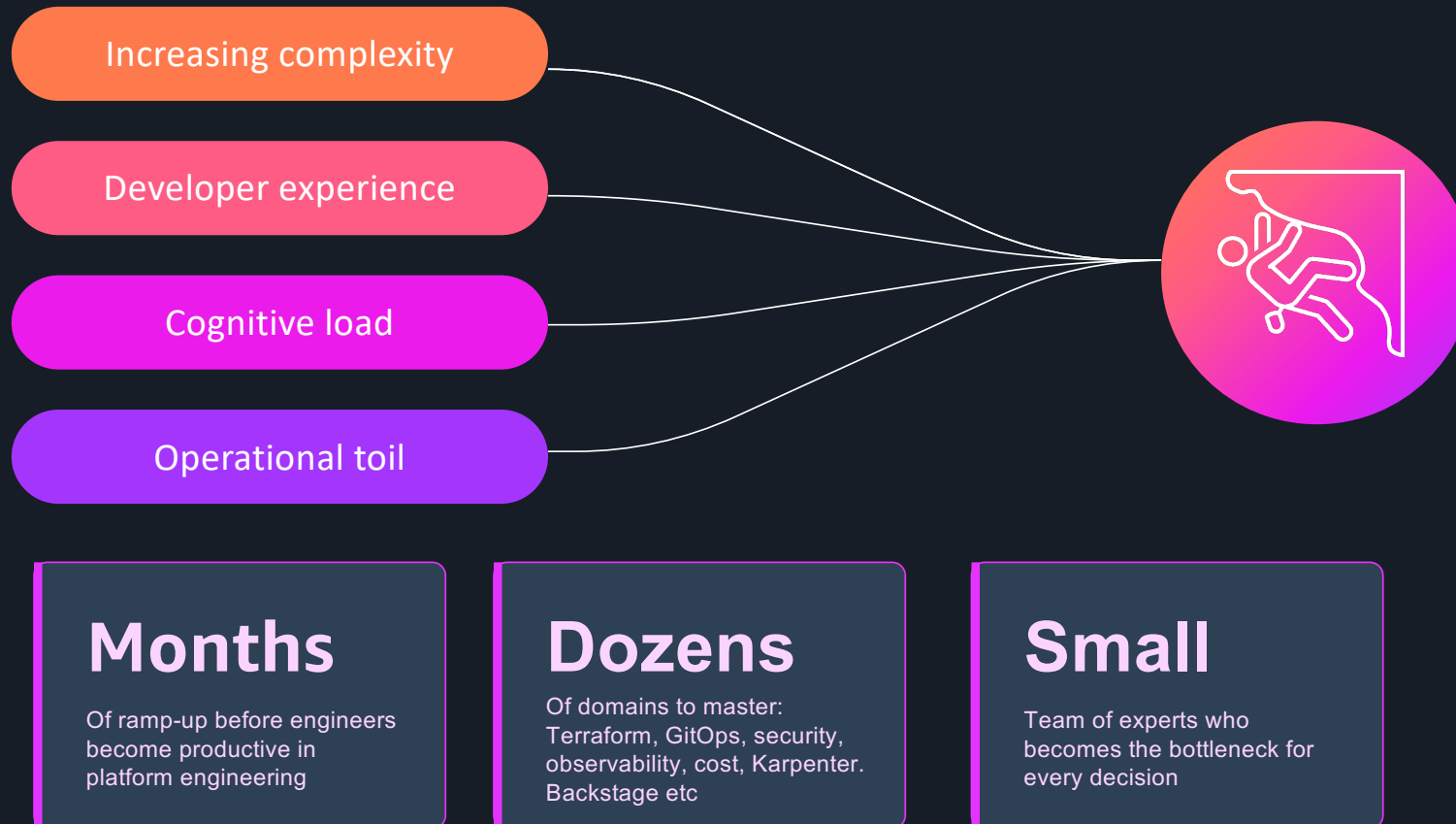
Agentic Platform Engineering Experience



APEX:

Agentic Platform Engineering eXperience

The platform engineering bottleneck



The platform engineering challenges



SPEED



TRUST



TALENT



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Agent Skills: Encoded expertise at scale

How **SKILL.md** provide steering for AI-generated output

SKILL.md – The professional step-by-step recipes

How Agent should do

- Decision frameworks (compute selection, networking)
- Terraform patterns and examples
- Security and compliance guardrails
- Operational best practices

Example: skill application flow

User prompt

"Design fully private EKS cluster with limited access"

eks-best-practices skill activated

Output follows AWS Well-Architected



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Introducing the EKS MCP Server

MCP tools – The professional kitchen

What Agent can do

- Read Kubernetes resources
- Get CloudWatch logs and metrics
- Apply configurations
- Describe EKS clusters and add-ons



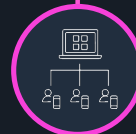
Kubernetes API

Allow LLMs to manage Kubernetes resources like Pods and Deployments



Troubleshooting

Access Kubernetes events and logs, search a knowledge base



Cluster management

Manage simple EKS Auto Mode clusters through CloudFormation

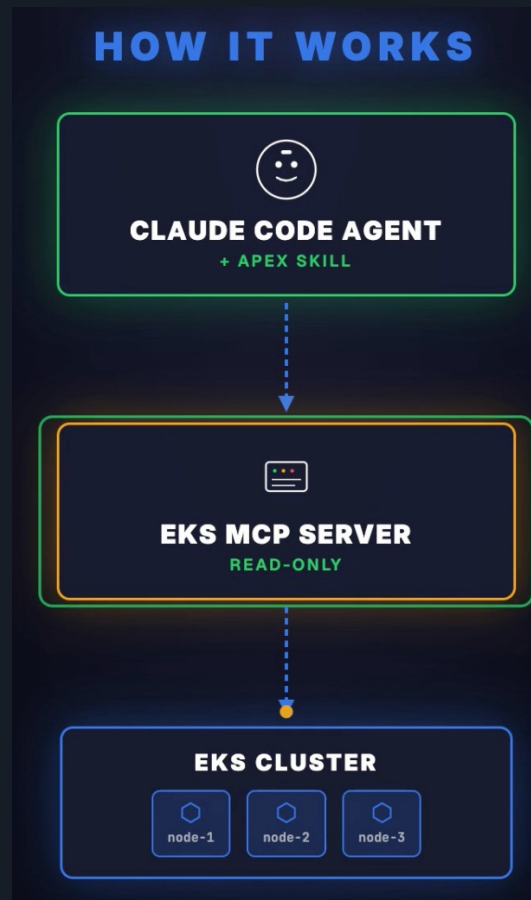


Deploy applications

Help LLMs apply Kubernetes YAML manifests



The flow



Key distinction: Skills **steer** agent behavior.

MCP tools connect to live infrastructure.

https://eks-mcp.{region}.api.aws/mcp

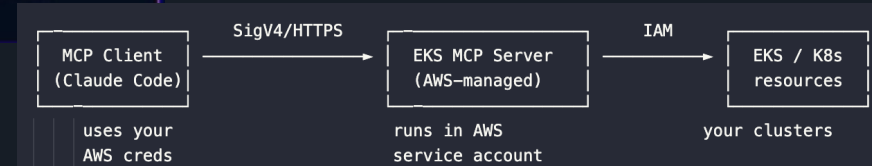
EKS MCP Server Tools

IAM AND SECURITY

```
"mcpServers": {
  "eks-mcp": {
    "type": "stdio",
    "command": "/opt/homebrew/bin/uvx",
    "args": [
      "mcp-proxy-for-aws@latest",
      "https://eks-mcp.ap-southeast-1.api.aws/mcp",
      "--service",
      "eks-mcp",
      "--profile",
      "default",
      "--region",
      "ap-southeast-1"
    ],
    "env": {
    }
  }
},
```

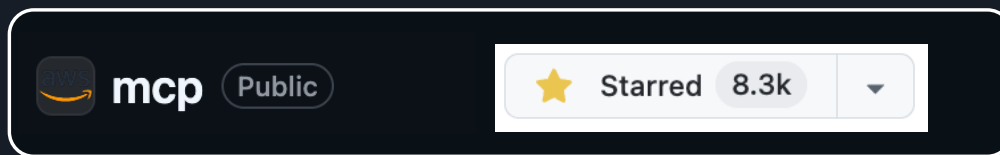
- Fully managed by AWS
- SigV4 authentication
- Open source

get_policies_for_role	• Retrieves all policies attached to an IAM role • Shows managed and inline policies with their documents
add_inline_policy	• Creates new inline policies and adds them to IAM roles • Supports single or multiple policy statements
get_cloudwatch_metrics	• Supports various resource types with flexible time ranges and aggregation
get_eks_metrics_guidance	• Provides information about available CloudWatch metrics for EKS resources • Helps determine correct dimensions for metric queries
get_k8s_events	• Retrieves Kubernetes events related to specific resources • Useful for troubleshooting pod failures and deployment issues
	• Creates both Deployment and LoadBalancer Service resources
describe_eks_resource	• Retrieves detailed information about a specific EKS cluster resource including configuration, status, and metadata.



MCP servers

<https://awslabs.github.io/mcp/>
<https://github.com/awslabs/mcp>



AWS API MCP
Server



Amazon EKS MCP
Server



AWS Lambda MCP
Server



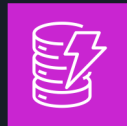
AWS Knowledge
MCP Server



AWS Step Functions
MCP Server



Amazon DocumentDB
MCP Server



Amazon DynamoDB
MCP Server



AWS Cost Explorer
MCP Server

Open source

55+ MCP servers

1M+ downloads



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Platform engineering life cycle



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Three Stages

Architect

- Business -> Tech Requirements
- Architecture Decisions

Build

- Infrastructure Provisioning
- Core capabilities

Operate

- Patching /upgrade
- Troubleshooting

DAY 0

Architect: Design with AI-guided decisions

What the agent does

- Design guidance and trade-off analysis
- Service selection recommendations
- Topology decisions (VPC, subnets, AZs)
- Compute option evaluation

Decisions guided by skills

- Karpenter vs MNG vs Auto Mode
- Networking patterns (VPC CNI modes)
- Security posture (Pod Identity, IRSA)
- Cost optimization strategies

EXAMPLE PROMPT

"Design a multi-tenant EKS platform for 20 developers with cost visibility"

SKILLS ACTIVATED

OUTPUT

Architecture Decision Record (ADR)



DAY 1

Build: Generate and deploy IaC with guardrails

What the agent does

- Transform architecture into Terraform
- Deploy EKS cluster and add-ons
- Configure GitOps tooling (Argo CD)
- Apply security scanning (eg: Checkov)

Built-in validation

- terraform fmt and validate
- Checkov security scans
- OPA policy checks
- Human review gates

EXAMPLE PROMPT

"Generate Terraform for a private EKS cluster with Karpenter"

SKILLS ACTIVATED

GENERATED TERRAFORM SNIPPET

```
module "eks" {  
  source = "terraform-aws-modules/eks/aws"  
  version = "~> 20.0"  
  
  cluster_name = "platform-eks"  
  cluster_version = "1.31"  
  
  cluster_endpoint_public_access = false  
  enable_irs = true  
}
```

OUTPUT

Working EKS platform via IaC



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DAY 2

Operate: EKS Upgrade Readiness and Operation Review

EKS Upgrade Readiness Skill

- EKS Upgrade Readiness Prep
 - Readiness Scoring
- Well Architected Review based on Best Practices

EKS Operation Review

- Cluster Lifecycle
- Observability
- Networking
- Workload

Readiness Score: 71% — FAIR

Your cluster has several issues that need attention before upgrading to 1.31. The two critical blockers are: (1) Karpenter v0.37.0 is incompatible with EKS 1.31 and

EKS Operation Review Report

Cluster: test-old-cluster | Region: us-east-1 | Version: 1.30 | Platform: eks.63

Date: 2026-04-02

Account: 00E*****

Executive Summary

test-old-cluster is an eksctl-managed EKS cluster running Kubernetes 1.30 on Extended Support, with 2 nodes across 2 AZs serving approximately 30 pods. The cluster demonstrates some good foundations — Karpenter is deployed with consolidation enabled, RBAC is appropriately scoped, Pod Disruption Budgets cover most workloads, CoreDNS is properly configured, and IP capacity is abundant.

However, the cluster has significant operational gaps across security and observability. The API server is publicly accessible to **0.0.0.0/0** with no private endpoint and all control plane logging disabled — meaning there is no audit trail and the API server is internet-exposed. The entire observability stack is absent: no metrics collection, no centralized logging, and no alerting. This means issues cannot be detected until users report them. The EBS CSI driver is in CrashLoopBackOff due to missing IAM credentials, impacting any EBS volume operations.

The most immediate cost concern is Extended Support: at \$0.60/hr vs \$0.10/hr, the cluster is paying approximately \$4,300/month extra. Upgrading to v1.32+ would eliminate this premium. Combined with the 13 RED findings (40.6% of rated items), this cluster requires urgent attention across security, observability, and version currency before it can be considered production-ready.

Maturity Score

Rating	Count	Percentage
GREEN	6	18.8%
AMBER	13	40.6%
RED	13	40.6%
UNKNOWN	5	--

Total items assessed: 37 (32 rated + 5 UNKNOWN)

HOW IT WORKS



CLAUDE CODE AGENT
+ APEX SKILL



EKS MCP SERVER
READ-ONLY



EKS CLUSTER



OUTPUT

EKS Operation Review and Actionable items



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DAY 2

Operate: Troubleshoot and optimize conversationally

What the agent does

- Diagnose operational issues
- Analyze costs and recommend savings
- Plan and validate upgrades
- Generate runbooks and documentation

Real-time capabilities

- Read pod status and events
- Query CloudWatch logs
- Inspect cluster configuration
- Check EKS Insights for issues

EXAMPLE PROMPTS

"Why are my pods stuck in Pending?"

"What cost optimizations can we make?"

"Prepare for EKS 1.31 upgrade"

MCP TOOLS IN ACTION

list_k8s_resources: Check pod status

get_cloudwatch_logs: Retrieve container logs

get_eks_insights: Identify cluster issues

OUTPUT

Operational runbooks + optimization actions



Start read-only. Expand with confidence.

Agent permission levels for building trust

1

Read-only

Agent queries only: list pods, get logs, describe resources. No mutations.

2

Review

Agent proposes changes; human manually applies. Engineer runs terraform apply or merges PR.

3

Gates

Agent triggers CI/CD; automated checks + human approval at checkpoints. ArgoCD manual sync.

4

Autonomous

Agent operates within guardrails. ArgoCD auto-sync + OPA policies + auto-rollback.

DORA 2025

"AI is an amplifier" - strong engineering practices see greater benefit

Adoption principles

- Start with read-only MCP tools
- Human-in-the-loop for mutations
- Expand scope as trust builds
- Guardrails replace approvals over time



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Resources

Workshop : <https://catalog.workshops.aws/apex-lab/en-US>

Repo: <https://s12d.com/APEX>

Site: <https://aws-samples.github.io/sample-apex-skills/>



AWS DevOps Agent

- Incident lifecycle and customer challenges
- Introducing AWS DevOps Agent
- Use cases
- Pricing
- Customer references



Incident Lifecycle & Challenges

Five Phases of Incident Management



AWS DevOps Agent

Drive operational excellence with an autonomous AI agent that resolves and proactively prevents incidents in AWS, multi-cloud, and on-prem environments

Autonomous incident response

Proactive incident prevention

On-demand SRE task handling



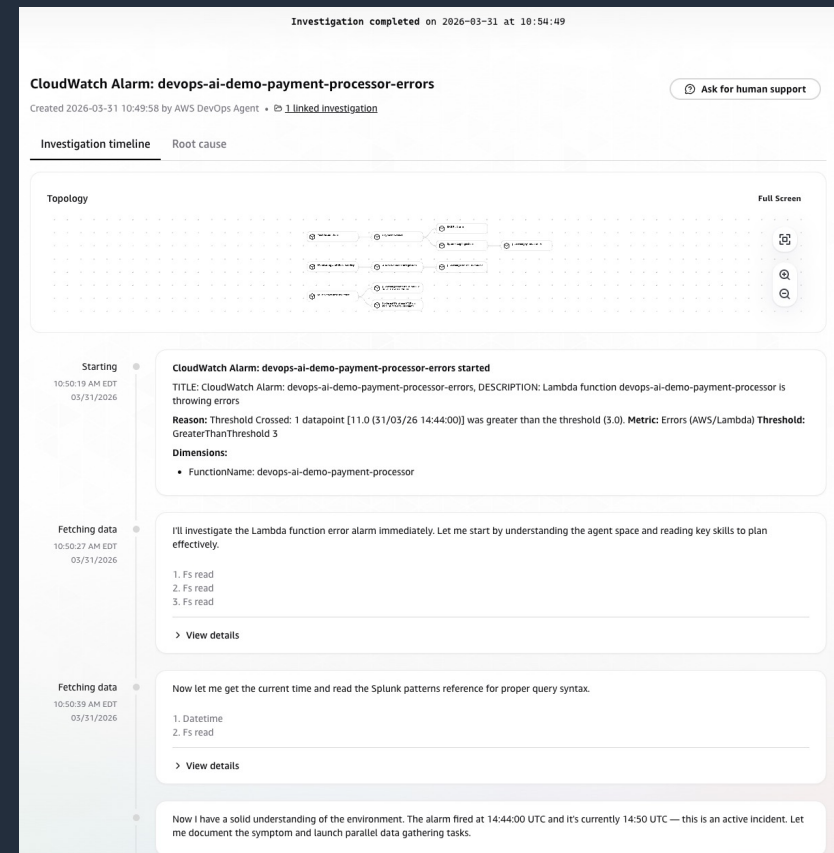
Autonomous incident response

Automatically investigates operational issues across AWS, multi-cloud, and on-prem environment

Correlates and analyzes telemetry, code, and deployment data to identify root causes

Provides detailed mitigation plans

Learns from past investigations to get faster and smarter over time



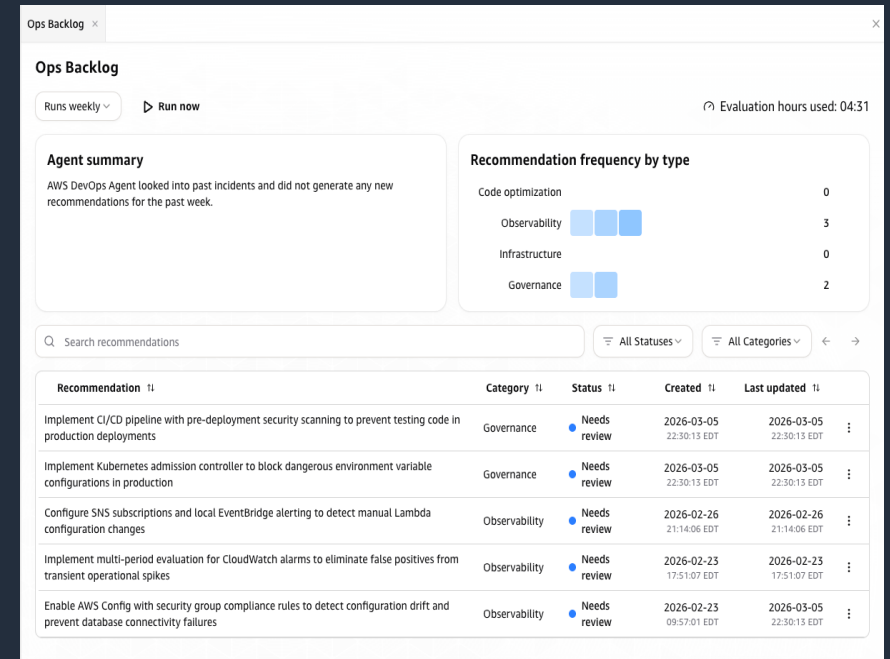
Proactive incident prevention

Move from reactive firefighting to proactive operational improvement

Analyzes historical incidents

Identifies gaps in observability coverage to reduce mean time to detect

Provides targeted recommendations to prevent future incidents and strengthen system resilience



On-demand SRE Task Handling

Get instant, contextual answers, and create shareable custom charts and reports

Built-in integrations with AWS, observability tools (Dynatrace, DataDog, NewRelic, Splunk), and CI/CD platforms (GitHub, GitLab, Azure DevOps)

Extensible via private or remote MCP servers

Automatically maps application relationships in a dynamic topology

Add reusable to execute tasks consistently and reliably

Do I have any certificates about to expire?

08:28 PM

Good news — **no certificates are expiring soon** because your infrastructure isn't currently using custom ACM certificates.

Here's what I found:

Certificate Status

Resource Type	Finding
ACM Certificates	None in use
IAM Server Certificates	None found
CloudFront	2 distributions using default Cloud
Load Balancers	All 6 ALBs use HTTP only — no SSL

💡 Something to Consider

Your load balancers are currently serving traffic over HTTP (port 80). While this can be fine for internal traffic behind CloudFront, you might want to consider adding HTTPS listeners with ACM certificates if these endpoints are publicly accessible — especially for the ShopSmart production services.

Would you like me to look into setting up SSL/TLS for your load balancers, or is there something else you'd like to check?

08:30 PM

Use Cases



Incident response and resolution

Autonomously triages incidents and guides teams to rapid resolution



Automated incident coordination

Works directly within your collaboration tools like ServiceNow and Slack to share findings and coordinate response



Proactively prevent future incidents

Analyzes patterns across historical incidents to deliver targeted recommendations that prevent future incidents and strengthen system resilience

AWS DevOps Agent: Transforming Incident Response and Operations

Metrics derived from 10+ Customer & Partner references



MTTR REDUCTION

75%

Average

Range: 50-92%



INVESTIGATION TIME

80%

Faster

42 min → 8 min per incident



ROOT CAUSE ACCURACY

94%

Average

Range: 84-100% accuracy



RESOLUTION SPEED IMPROVEMENT

8.5x

Faster

Range: 7-10x faster



PRODUCTIVITY GAINS

57 hrs/mo

Average Hours/Month Saved

Range: 22-120 hrs/month saved



Detailed customer and partner quotes



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Thank you